Akoizo Original Practice - 2027 Rotation Form 1 - Answer key

Division B | Food Science | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Lactose is made of which two sugars?

Unit: What milk is made of: sugars, fats, and proteins.
Worked-case data: A 250 mL glass of whole milk weighs about 258 g. Using 3.3% protein and 4.8% lactose by mass, estimate the grams of protein and lactose.
Separate investigation observations:
Liquid A: Clear after acid is added and curds are removed.
Test of A: Contains lactose and dissolved proteins.
Solid B: White clumps that formed at pH 4.6.

A. Glucose and fructose

B. Two glucose units

C. Galactose and fructose

D. Glucose and galactose

Correct: D

2. (2 points) Casein clumps most at about which pH?

Unit: Proteins and curds.
Worked-case data: Milk in a beaker has mass 245.0 g; the empty beaker is 120.8 g. After curdling and draining, the curds on a 3.2 g cheesecloth weigh 22.0 g together. Find percent curd.
Separate investigation observations:
Trial A: Whole milk, 2 mL vinegar: 15% curd.
Trial B: Skim milk, 2 mL vinegar: 11% curd.
Trial C: Whole milk, 0.2 mL vinegar: 3% curd, cloudy whey.

A. 4.6

B. 6.7

C. 7.4

D. 9.0

Correct: A

3. (2 points) HTST pasteurization uses:

Unit: Processing, cooking, and preservation.
Worked-case data: A dairy heats milk to 72 °C for 15 seconds, homogenizes it, and stores it at 4 °C. Another heats to 140 °C for 3 seconds and packages it in sterile cartons. Compare the products.
Separate investigation observations:
Sample X: Raw milk left at room temperature for 2 days: sour smell, clumps.
Sample Y: HTST milk, refrigerated 2 days: fresh.
Sample Z: Unopened UHT carton at room temperature, 2 months: fresh.

A. 140 °C for 30 minutes

B. 72 °C for 15 seconds

C. 63 °C for 15 seconds

D. 100 °C for 1 hour

Correct: B

4. (2 points) Which product is safe for someone with lactose intolerance but NOT for someone with milk allergy?

Unit: Lactose intolerance, allergies, alternatives, and the notebook.
Worked-case data: A student plans to test how vinegar volume affects curd mass. Design one notebook experiment.
Separate investigation observations:
Person 1: Bloating after milk; fine with yogurt and lactose-free milk.
Person 2: Hives within minutes of a tiny amount of cheese.
Test: Person 2 has antibodies to casein.

A. Water

B. Rice milk

C. Lactose-free cow’s milk

D. Oat milk

Correct: C

5. (2 points) Forming one triglyceride releases how many water molecules?

Unit: Fats: saturation, esters, and emulsions.
Worked-case data: A team churns 150 mL of cream (mass 148 g) and collects 52 g of butter. If butter is about 80% fat, estimate the cream's fat percentage.
Separate investigation observations:
Trial 1: Cream at 4 °C: no butter after 15 minutes of shaking.
Trial 2: Cream at 13 °C: butter formed in 8 minutes.
Trial 3: Cream at 25 °C: soft, greasy butter; cloudy buttermilk.

A. None

B. Three

C. One

D. Two

Correct: B

6. (2 points) Which solute lowers freezing point more per mole?

Unit: Freezing point depression and ice cream.
Worked-case data: Estimate the freezing point of meltwater in which 2.0 mol of NaCl dissolves per kilogram of water.
Separate investigation observations:
Trial 1: Milk, 30 mL salt: soft slush, 2 cm tall.
Trial 2: Milk, 90 mL salt: firmer, 4 cm tall.
Trial 3: Cream, 90 mL salt: firm and smooth, 7 cm tall.

A. They are equal

B. Neither

C. NaCl

D. Sugar (sucrose)

Correct: C

7. (2 points) On a label, 25% DV of a nutrient is considered:

Unit: Nutrition and food labels.
Worked-case data: A label lists 300 mg of calcium per serving and a Daily Value of 1,300 mg. A student drinks 3 servings. What percent DV is that, and is one serving “high”?
Separate investigation observations:
Milk A: 8 g fat, 12 g total sugars, 0 g added sugars, 25% DV calcium.
Milk B: 0 g fat, 12 g total sugars, 0 g added sugars, 25% DV calcium.
Milk C: 5 g fat, 24 g total sugars, 12 g added sugars, 25% DV calcium.

A. Low

B. Moderate only

C. Not meaningful

D. High

Correct: D

8. (2 points) A triglyceride contains glycerol and:

Unit: What milk is made of: sugars, fats, and proteins.
Worked-case data: A 250 mL glass of whole milk weighs about 258 g. Using 3.3% protein and 4.8% lactose by mass, estimate the grams of protein and lactose.
Separate investigation observations:
Liquid A: Clear after acid is added and curds are removed.
Test of A: Contains lactose and dissolved proteins.
Solid B: White clumps that formed at pH 4.6.

A. Two glucose units

B. One phosphate and one sugar

C. Three fatty acids

D. Three amino acids

Correct: C

9. (2 points) Why can boiling fresh milk not cause curdling by itself?

Unit: Proteins and curds.
Worked-case data: Milk in a beaker has mass 245.0 g; the empty beaker is 120.8 g. After curdling and draining, the curds on a 3.2 g cheesecloth weigh 22.0 g together. Find percent curd.
Separate investigation observations:
Trial A: Whole milk, 2 mL vinegar: 15% curd.
Trial B: Skim milk, 2 mL vinegar: 11% curd.
Trial C: Whole milk, 0.2 mL vinegar: 3% curd, cloudy whey.

A. Whey proteins cannot denature

B. Boiling raises pH to 10

C. Lactose prevents clumping

D. Casein is heat-stable at normal pH

Correct: D

10. (2 points) Why does yogurt thicken?

Unit: Processing, cooking, and preservation.
Worked-case data: A dairy heats milk to 72 °C for 15 seconds, homogenizes it, and stores it at 4 °C. Another heats to 140 °C for 3 seconds and packages it in sterile cartons. Compare the products.
Separate investigation observations:
Sample X: Raw milk left at room temperature for 2 days: sour smell, clumps.
Sample Y: HTST milk, refrigerated 2 days: fresh.
Sample Z: Unopened UHT carton at room temperature, 2 months: fresh.

A. Lactic acid lowers pH and casein gels

B. Fat crystallizes

C. Sugar is added

D. Water evaporates

Correct: A

11. (2 points) A dairy label gives 9 g carbohydrate, 4 g protein and 3 g fat per serving. Using 4/4/9 kcal per gram, calculate energy.

answer: 79 kcal (2 points).

Numerical target 79; absolute tolerance 0.158. Accepted units: , kcal.

12. (2 points) A milk experiment begins with 110 g milk and produces 11 g drained curd. Find curd mass as percent of starting milk mass.

answer: 10 % (2 points).

Numerical target 10; absolute tolerance 0.02. Accepted units: , %.

13. (2 points) A cream experiment starts with 110 g cream and produces 22 g butter. Find butter mass yield percentage.

answer: 20 % (2 points).

Numerical target 20; absolute tolerance 0.04. Accepted units: , %.

14. (2 points) A team obtains 3 g curd while the supervisor obtains twice that mass. Using mass ratio times 10, calculate the task score.

answer: 5 (2 points).

Numerical target 5; absolute tolerance 0.01. Accepted units: , .

15. (2 points) A synthetic ice-cream mound stands 90 mm high for the specified five seconds. Using 0.1 points/mm, calculate its height score.

answer: 9 (2 points).

Numerical target 9; absolute tolerance 0.018. Accepted units: , .

16. (2 points) Explain why adding salt to the surrounding ice bath helps cool a sealed dairy mixture.

answer: Dissolved salt lowers the freezing point of the external water/ice mixture, allowing a colder bath and heat transfer out of the sealed dairy mixture. (2 points).

17. (2 points) Explain how acid addition can cause casein-rich curds to separate from milk.

answer: Acid lowers pH toward conditions that reduce casein-micelle stability, promoting aggregation; the liquid whey can then be drained from the curds. (2 points).

18. (2 points) Distinguish lactose intolerance from a milk-protein allergy.

answer: Lactose intolerance concerns inadequate digestion of lactose by lactase; milk-protein allergy involves an immune response to milk proteins. They are different mechanisms. (2 points).

19. (2 points) Does lactase treatment remove all carbohydrate from milk? Explain.

answer: No. Lactase hydrolyzes lactose into smaller sugars; it does not simply remove the sugar mass. (2 points).

20. (2 points) Why must a dairy-curd comparison standardize draining time and starting milk quantity?

answer: Retained liquid and different initial sample amounts can change measured curd mass independently of the experimental variable. (2 points).